11(McKEOWN et al., 2008) 110(XIA et al., 2015) 111(HAN et al., 2015) 112(JAIN et al., 2013) 113(KOPONEN et al., 2014) 114(MEDVED et al., 2014) 115(BERDE et al., 2014) 116(NTT Laboratories. Ryu SDN Framework - Open-source SDN Platform Software. NTT Technical Review, 2014) 117(NOXRepo. POX: The POX network software platform (controlador OpenFlow em Python). Projeto open-source (sucessor educacional do NOX)., ) 118(HELLER et al., 2012) 119(YEGANEH et al., 2012) 12(GUDE et al., 2008) 120(CURTIS et al., 2011) 121(YU et al., 2010) 122(PORRAS et al., 2012) 123(LI et al., 2016) 124(BHUIYAN et al., 2023) 125(WANG et al., 2021) 126(LATIF et al., 2022) 127(CHIOSI et al., 2012) 128(ETSI NFV ISG. GS NFV 002 V1.1.1: Network Functions Virtualisation (NFV); Architectural Framework. ETSI et al., 2013) 129(ETSI NFV ISG. GS NFV-MAN 001 V1.1.1: Network Functions Virtualisation (NFV); Management and Orchestration. ETSI et al., 2014) 13(CASADO et al., 2007) 130(MIJUMBI et al., 2015) 131(MIJUMBI et al., 2016) 132(LAGHRISSI et al., 2019) 133(LEIVADEAS et al., 2022) 134(MORALES et al., 2017) 135(VELASCO et al., 2022) 14(LANTZ et al., 2010) 15(PFAFF et al., 2015) 16(SHERWOOD et al., 2009) 17(Open Networking Foundation (ONF). OpenFlow Switch Specification, 2009) 18(FEAMSTER et al., 2013) 19(KREUTZ et al., 2015) 21(BRUTLAG, 2000) 210(BHUYAN et al., 2014) 211(GARCÍA-TEODORO et al., 2009) 212(BUCZAK et al., 2016) 213(CORTES et al., 1995) 214(SCHÖLKOPF et al., 2001) 215(MUKKAMALA et al., 2003) 216(BREIMAN, 2001) 217(CARUANA et al., 2006) 218(TAVALLAEE et al., 2009) 219(STOLFO et al., 2000) 22(ROBERTS, 1959) 220(MOUSTAFA et al., 2015) 221(SHARAFALDIN et al., 2018) 222(HODO et al., 2017) 223(FARNAAZ et al., 2016) 224(ALDUAILIJ et al., 2022) 225(NEGANDHI et al., 2016) 226(WILLEMS et al., 2023) 227(SHEHAB et al., 2024) 228(ALMAIAH et al., 2025) 229(STANIFORD-CHEN et al., 1996) 23(ZHANG et al., 2020) 230(LEE et al., ) 231(CANNADY, 1998) 232(LAKHINA et al., 2004) 233(AHMED et al., 2007) 234(FERNANDES et al., 2019) 235(SCHUMMER et al., 2024) 236(BENKŐ et al., 2020) 24(ABOODE, 2018) 25(MOAYEDI et al., 2008) 26(PENA et al., ) 27(FONDO-FERREIRO et al., 2026) 28(CHANDOLA et al., 2009) 29(AHMED et al., 2016) 31(LIU et al., 2025) 310(TOUVRON et al., 2023) 311(TOUVRON et al., 2023) 312(JIANG et al., 2023) 313(ALMAZROUEI et al., 2023) 314(BAI et al., 2023) 315(YANG et al., 2024) 316(BIGSCIENCE WORKSHOP et al., 2022) 317(GEMMA TEAM (Google DeepMind). Gemma: Open Models Based on Gemini Research and Technology. arXiv preprint arXiv:2403.08295, 2024) 318(HU et al., 2022) 319(VASWANI et al., 2017) 32(LIU et al., 2024) 320(OUYANG et al., 2022) 321(WEI et al., 2022) 322(CHEN et al., 2024) 323(WANG et al., 2024) 324(HOSSAIN et al., 2025) 325(SDN-Xplain: LLM-Driven Anomaly Detection and Human-Interpretable

Explanation in SDN. SKKU IoT Lab / KICS Winter 2026 (preprint)., 2026) 326(Evaluating Language Models for Threat Detection in IoT Security Logs. arXiv:2507.02390, 2025) 327(Lightweight Fine-Tuning of LLMs for Explainable Intrusion Detection in SDN. arXiv/ResearchGate preprint, 2025) 328(Attribution-Driven Explainable Intrusion Detection with Encoder-based LLM for SDN. arXiv:2604.06266, 2026) 329(Large Language Models powered Malicious Traffic Detection. arXiv:2503.18487, 2025) 33(BOATENG et al., 2024) 330(SENTINEL: A Semantic-Aware Network Intelligence Framework ... (LLM em anomaly detection convertendo logs em linguagem natural). IEEE TrustCom 2025., 2025) 331(ALSAMARNEH et al., 2025) 332(ADANZA et al., 2025) 333(Towards Intent-Based Network Management: Large Language Models for Intent Extraction in 5G Core Networks. IEEE / GenAI Net research library, 2024) 334(Exploring LLM-Based Agents for Root Cause Analysis. ACM SIGMETRICS / arXiv:2403.04123, 2024) 335(Empowering AIOps: Leveraging Large Language Models for IT Operations Management. arXiv:2501.12461, 2025) 336(WANG et al., 2025) 337(LLM-Driven Multi-Agent Framework for Autonomous Network Slice Management. IEEE, 2026) 338(CHEN et al., 2024) 339(LIU et al., 2025) 34(HANG et al., 2024) 340(Large Language Models Meet Next-Generation Networking Technologies: A Review (vide [4]) também cobre segurança de LLM em redes. (Incluído por overlaps temáticos; contagem principal permanece em [4].) ==========
et al., ) 35(HUANG et al., 2024) 36(HONG et al., 2025) 37(ZHANG et al., 2025) 38(Large Language Models for Agentic NetOps and AIOps (Survey). arXiv:2605.12729, 2026) 39(TANG et al., 2024) 41(HU et al., 2021) 410(BENZAKEN et al., 2022) 411(HU et al., 2023) 412(ZHANG et al., 2024) 413(HE et al., 2022) 414(DING et al., 2022) 415(HAN et al., 2024) 416(ZHANG et al., 2025) 417(SLAMANIG et al., 2025) 418(TOUVRON et al., 2023) 419(TOUVRON et al., 2023) 42(DETTMERS et al., 2023) 420(JIANG et al., 2023) 421(GRATTAFIORI et al., 2024) 422(BROWN et al., 2005) 423(WEI et al., 2022) 424(ACHIAM et al., 2023) 425(MENG et al., 2025) 426(GINIGE et al., 2024) 427(CUI et al., 2025) 428(PENG et al., 2024) 429(GRUVER et al., 2023) 43(ZHANG et al., 2023) 430(SCHWARTZ et al., 2024) 431(SUI et al., 2024) 432(HEGSELMANN et al., 2023) 433(JIANG et al., 2025) 434(ALNEGHEIMISH et al., 2024) 435(ZHANG et al., 2024) 436(CAO et al., 2024) 437(KUNDU et al., 2024) 44(LIU et al., 2024) 45(HOULSBY et al., 1902) 46(LI et al., 2021) 47(LIU et al., 2022) 48(LESTER et al., 2021) 49(LIU et al., 2022) 51(ESSENCIAL Danilevsky et al., 2010) 510(RECOMENDADO Montavon et al., 2018) 511(ESSENCIAL Ali et al., 2023) 512(ESSENCIAL Neupane et al., 2022) 513(RECOMENDADO Amarasinghe et al., 2018) 514(RECOMENDADO Kenney et al., 2022) 515(RECOMENDADO Gaspar et al., 2024) 516(RECOMENDADO Rjoub et al., 2023) 517(RECOMENDADO

Sarker, 2024) 518(COMPLEMENTAR Sharma et al., 2025) 519(COMPLEMENTAR Sivakolundhu, 2022) 52(ESSENCIAL Arrieta et al., 2019) 520(COMPLEMENTAR Houssel et al., 2024) 521(COMPLEMENTAR Pizarro et al., 2025) 522(COMPLEMENTAR Lightweight Fine-Tuning of LLMs for Explainable Intrusion Detection in SDN. arXiv:2510.23313 (preprint), 2025) 523(COMPLEMENTAR Khan et al., 2025) 524(COMPLEMENTAR Explainable AI for Network Intrusion Detection Systems: A Survey and Open Research Questions. arXiv preprint et al., 2023) 525(ESSENCIAL Chen et al., 2024) 526(RECOMENDADO Zhang et al., 2025) 527(RECOMENDADO Yamin et al., 2024) 528(RECOMENDADO Habibzadeh et al., 2026) 529(RECOMENDADO LLMs in the SOC: An Empirical Study of Human-AI Collaboration. arXiv:2508.18947, 2025) 53(ESSENCIAL Ribeiro et al., 2016) 530(COMPLEMENTAR Hasanov, 2025) 531(ESSENCIAL Kadavath et al., 2022) 532(RECOMENDADO Guo et al., 2017) 533(RECOMENDADO Lin et al., 2022) 534(RECOMENDADO Zhu et al., 2023) 535(COMPLEMENTAR Benchmarking Uncertainty Estimation in Large Language Models. OpenReview / TMLR (submissão). [referência de calibração/uncertainty de LLM], ) 536(COMPLEMENTAR Li et al., 2025) 537(ESSENCIAL Wei et al., 2022) 538(RECOMENDADO Dang et al., 2024) 54(ESSENCIAL Lundberg et al., 2017) 55(RECOMENDADO Adadi et al., 2018) 56(RECOMENDADO Guidotti et al., 2018) 57(RECOMENDADO Molnar, 2022) 58(RECOMENDADO Gilpin et al., 2018) 59(RECOMENDADO Samek et al., 2019) 61(ESSENCIAL - Dataset UNSW-NB15 (descrição oficial / paper fundador) Moustafa et al., 2015) 610(RECOMENDADO - Dataset InSDN et al., 2020) 611(RECOMENDADO - NetFlow datasets et al., 2011) 612(RECOMENDADO - Review de datasets de IDS Thakkar et al., 2020) 613(COMPLEMENTAR - Deep learning para NIDS Javaid et al., 2016) 614(RECOMENDADO - Data augmentation em IoT (desbalanceamento) Zhang et al., 2022) 615(RECOMENDADO - Geração sintética e privacy-preserving de traces Aceto et al., 2024) 616(ESSENCIAL - AnoLLM et al., 2025) 617(RECOMENDADO - Survey LLMs em dados tabulares Fang et al., 2024) 618(RECOMENDADO - Detecção zero-shot de anomalias tabulares com LLMs Li et al., 2024) 619(COMPLEMENTAR - Dataset CIC-DDoS2019 Canadian Institute for Cybersecurity. CIC-DDoS2019 Dataset., ) 62(ESSENCIAL - Avaliação estatística do UNSW-NB15 Moustafa et al., 2016) 620(COMPLEMENTAR - Survey de emuladores/simuladores de rede Engelen et al., 2023) 621(RECOMENDADO - Geração de dados DDoS em SDN (Mininet+Ryu) Gayantha, 2025) 622(COMPLEMENTAR - Dataset SNT (Mininet+Ryu) Simulated Network Traffic Using Mininet and Ryu (SNT) Dataset. Mendeley Data., ) 623(RECOMENDADO - Augmentation GAN para classes minoritárias em rede Kang et al., 2025) 624(RECOMENDADO - Review de IDS

por ML/DL em IoT Asharf et al., 2020) 625(COMPLEMENTAR - Benchmark em NSL-KDD/CICIDS2017/UNSW-NB15 Waghmode et al., 2025) 626(COMPLEMENTAR - Análise KDD99/NSL-KDD/UNSW-NB15 Choudhary et al., 2021) 627(COMPLEMENTAR - PNETLab (emulador multi-vendor) EveNG / PNETLab. PNETLab - Lab is Simple (network emulator platform)., ) 628(COMPLEMENTAR - GNS3 (emulador gráfico) GNS3 Technologies. GNS3 - Graphical Network Simulator 3., ) 629(COMPLEMENTAR - Testbed híbrido SDN et al., 2023) 63(ESSENCIAL - Metodologia de geração CIC (base do CSE-CIC-IDS2018) Sharafaldin et al., 2018) 630(RECOMENDADO - Survey de geração sintética de tráfego de rede Synthetic Network Traffic Data Generation: A Comparative Analysis. arXiv preprint, 2024) 64(ESSENCIAL - Dataset CSE-CIC-IDS2018 (descrição oficial) Canadian Institute for Cybersecurity (CIC) & Communications Security Establishment (CSE). CSE-CIC-IDS2018 Dataset. University of New Brunswick., ) 65(ESSENCIAL - Survey de datasets de NIDS Ring et al., 2019) 66(RECOMENDADO - Análise do KDD Cup 99 / NSL-KDD et al., 2009) 67(RECOMENDADO - Dataset BoT-IoT et al., 2019) 68(RECOMENDADO - Ferramenta CICFlowMeter (geração de features) Canadian Institute for Cybersecurity. CICFlowMeter - Network Traffic Flow Generator and Analyzer., ) 69(ESSENCIAL - Mininet (emulador para SDN) Lantz et al., 2010) x 39 test 116.